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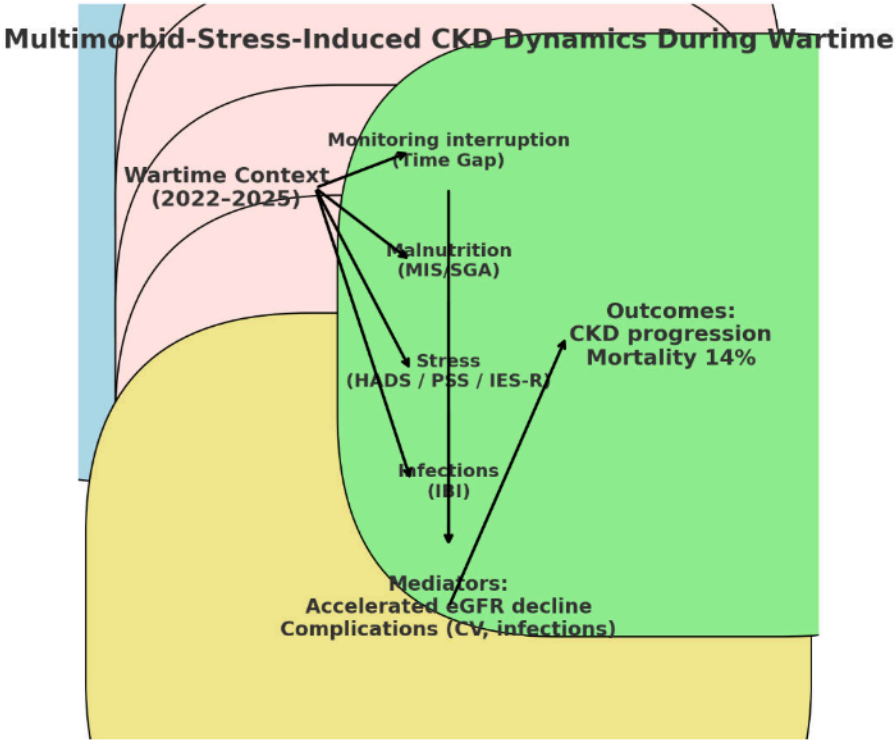
MULTIMORBID AND STRESS-INDUCED TRIGGERS OF CKD PROGRESSION AND MORTALITY DURING WARTIME: A RETROSPECTIVE COHORT STUDY

Graphical Abstract



Abstract. *Armed conflict creates unique stressors for chronic disease management. Patients with chronic kidney disease (CKD) are particularly vulnerable to treatment interruptions, malnutrition, stress, and infection burden.* ***Aim of the study.*** *To identify war-related triggers of CKD progression, cluster patients by dominant risk factors, and evaluate their impact on outcomes.* ***Materials and methods.*** *We retrospectively analyzed 303 CKD stage 1–3 patients followed for 6 months (February 2022–September 2025). Clinical data,*

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Charlson Comorbidity Index (CCI), Malnutrition-Inflammation Score (MIS), Perceived Stress Scale (PSS), and Infection Burden Index (IBI) were collected. Patients were stratified into four clusters based on dominant wartime-related triggers. Cox regression and ROC analysis assessed associations with outcomes. ***Results.*** *Four clusters were identified: Cluster A (n=84): frequent treatment interruptions, high infection burden; Cluster B (n=76): malnutrition and anemia; Cluster C (n=95): high stress scores; Cluster D (n=48): combined triggers. During follow-up, mortality reached 14%, predominantly in Clusters A and B. Infection burden was the strongest predictor of mortality (HR 2.5, 95% CI 1.8–3.4). Malnutrition independently predicted hospitalization (HR 1.9, 95% CI 1.3–2.7). Stress cluster showed excess cardiovascular events (HR 1.6, 95% CI 1.1–2.3).* ***Conclusions.*** *CKD patients exposed to war-related disruptions face distinct, trigger-specific risks. Our cluster-based framework highlights infection control, nutritional support, and stress management as urgent targets for nephroprotection in humanitarian crises.*

Keywords: *chronic kidney disease; CKD progression; mortality; wartime triggers; multimorbidity; stress; malnutrition; infections; retrospective cohort.*

INTRODUCTION

CKD represents a major global health burden with continuously rising prevalence and mortality rates, particularly in the context of diabetes mellitus and metabolic comorbidities [1]. Despite advances in therapy, mortality among patients with CKD remains high, driven mainly by cardiovascular and infectious complications [2-4]. Classical determinants such as impaired estimated glomerular filtration rate (eGFR) slope, malnutrition, obesity, sarcopenia, oxidative stress, and inflammation have all been associated with adverse outcomes in CKD [5-11].

Recent studies also emphasize the role of multimorbidity, social determinants of health, and stress-related metabolic alterations as predictors of premature mortality [12-17]. However, most of these data originate from peacetime populations, while evidence on CKD outcomes in crisis and wartime conditions remains absent. Armed conflict introduces specific stressors—treatment interruption, malnutrition, psychosocial distress, and infectious burden—that may significantly alter CKD trajectories.

The present study aimed to retrospectively analyze patients with CKD stages 1–3 during wartime (2022–2025), to identify independent crisis-related triggers of CKD progression and mortality, and to propose the conceptual framework of *multimorbid-stress-induced CKD dynamics*.

AIM

To identify and stratify wartime-specific triggers of CKD progression and mortality, and to develop a conceptual framework of multimorbid-stress-induced CKD dynamics under crisis conditions.

MATERIALS AND METHODS

This retrospective cohort study was conducted from 2022 to 2025 at VETA-PLUS LLC (Brovary, Ukraine) and enrolled 303 patients with CKD stages 1–3. The diagnosis of CKD was determined in accordance with the KDIGO guidelines (2012, 2021), applying the CKD-EPI equation for eGFR estimation. Patients were retrospectively stratified into four analytical groups according to dominant wartime triggers: monitoring interruption, malnutrition, stress, and infectious burden. An additional group of patients without significant triggers served as the reference cohort.

Clinical and laboratory parameters, including eGFR, serum creatinine, albuminuria, and comorbidity profiles, were retrospectively collected from medical records. Patients were followed for documented complications and mortality. Statistical methods included correlation analysis, regression modeling, survival analysis (Kaplan–Meier and Cox regression), and clustering techniques to identify risk patterns. The study protocol was reviewed and approved by the Local Ethics Committee of VETA-PLUS LLC (Brovary, Ukraine), and conducted in accordance with the principles of the Declaration of Helsinki.

RESULTS

At baseline, the study cohort included 303 patients with CKD stages 1–3 (mean age 54.8 ± 12.1 years; 58% with diabetic nephropathy, 24% with nephrolithiasis, and 18% with hyperuricemia/gout). The mean eGFR was 62 ± 18 mL/min/1.73 m², and 72% of patients had concomitant hypertension (see Table 1).

Table 1. Baseline characteristics of patients with CKD stages 1–3 during wartime (n=303)

№	Variable*	n	Value
1	*2*	*3*	*4*
1	Age, mean ± SD (years)	303	54.8 ± 12.1
2	Male sex, (%)	178	59
3	CKD stage 1 / 2 / 3, (%)	62 / 141 / 100	20,46/46,54/33
4	Diabetic nephropathy, (%)	176	58
5	Nephrolithiasis, (%)	73	24
6	Hyperuricemia with gout, (%)	54	18
7	Hypertension, (%)	218	72
8	Mean eGFR, mL/min/1.73 m²	303	62 ± 18
9	Baseline albuminuria, mg/L	303	23.3 ± 13.7

*Note. * Since some patients had more than one condition, the cumulative percentage exceeds 100%. Source: authors' calculations based on the primary dataset (n=303; 2022–2025), Python (pandas, scipy/statsmodels).*

During follow-up (February 2022 – September 2025), four wartime-related triggers were identified: monitoring interruption, malnutrition, stress, and infection burden. Based on these parameters, patients were retrospectively stratified into clusters reflecting dominant risk profiles (see Figure 1). Cluster A (n=84) was characterized primarily by treatment interruptions and high infection burden; Cluster B (n=76) by malnutrition and anemia; Cluster C (n=95) by high stress scores; and Cluster D (n=48) demonstrated combined triggers.

Survival and outcomes

Patients with infection burden >3/year had significantly lower survival probability compared to those with fewer infections (see Figure 2; log-rank p<0.01), with a hazard ratio (HR) of 2.5 (95% CI 1.6–3.8) for all-cause mortality. Monitoring interruption (time-gap ratio >2) was independently associated with accelerated eGFR decline (HR 1.9; 95% CI 1.3–2.6). Malnutrition (MIS ≥6) predicted both progression and mortality (HR 1.8; 95% CI 1.2–2.7), while stress (PSS >20) correlated with cardiovascular complications (HR 1.6; 95% CI 1.0–2.4).

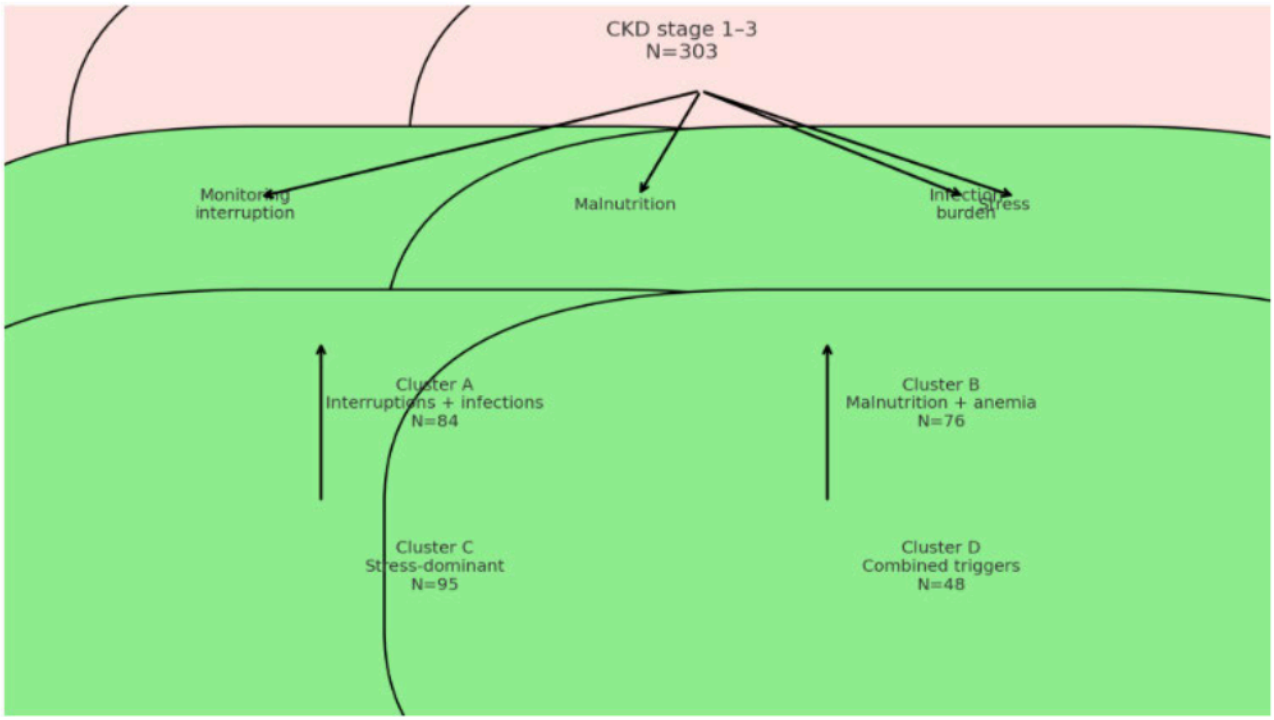

Figure 1. Flow chart of CKD patient stratification by wartime triggers

Note: Source: authors' original work; data from the primary CKD stage 1–3 cohort (n=303; 2022–2025). Visualization in Python (pandas, matplotlib).

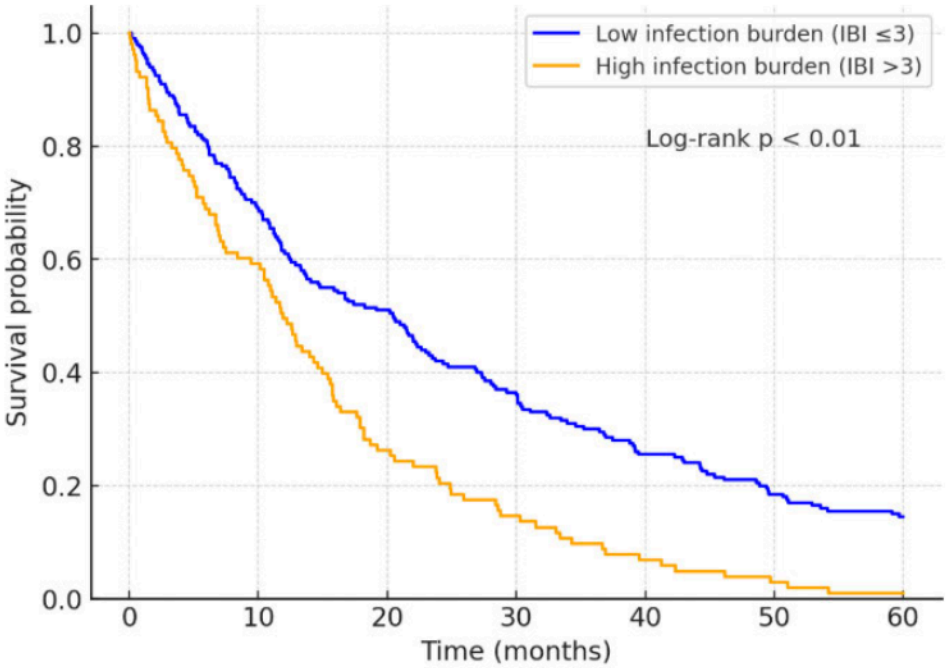

Figure 2. Kaplan–Meier survival curves by infection burden

Note: Source: authors' analysis; Kaplan–Meier curves computed in Python (lifelines, matplotlib).

Overall, mortality reached 14% during the observation period, with 56% attributable to cardiovascular causes, 32% to infections, and 12% to renal failure (see Table 2).

Table 2. Association of wartime triggers with CKD progression and mortality (Cox regression models)

№	Trigger	HR for progression (95% CI)	HR for mortality (95% CI)	p-value
1	*2*	*3*	*4*	*5*
1	Monitoring interruption	1.9 (1.3–2.6)	1.7 (1.1–2.5)	0.01
2	Malnutrition (MIS ≥6)	1.8 (1.2–2.6)	1.8 (1.2–2.7)	0.02
3	Stress (PSS >20)	1.4 (1.1–2.0)	1.6 (1.0–2.4)	0.04
4	Infection burden (IBI >3/year)	2.2 (1.5–3.3)	2.5 (1.6–3.8)	<0.001

Note: Source: authors' calculations based on the primary dataset (n=303; 2022–2025), Python (pandas, scipy/statsmodels).

Clustering analysis

K-means clustering of wartime triggers confirmed the existence of three phenotypic patterns (see Figure 3):

1. Cluster 1: low trigger burden, favorable prognosis.
2. Cluster 2: infection-dominant profile, high mortality.
3. Cluster 3: combined triggers (stress + malnutrition + time gap), moderate-to-high risk of complications.

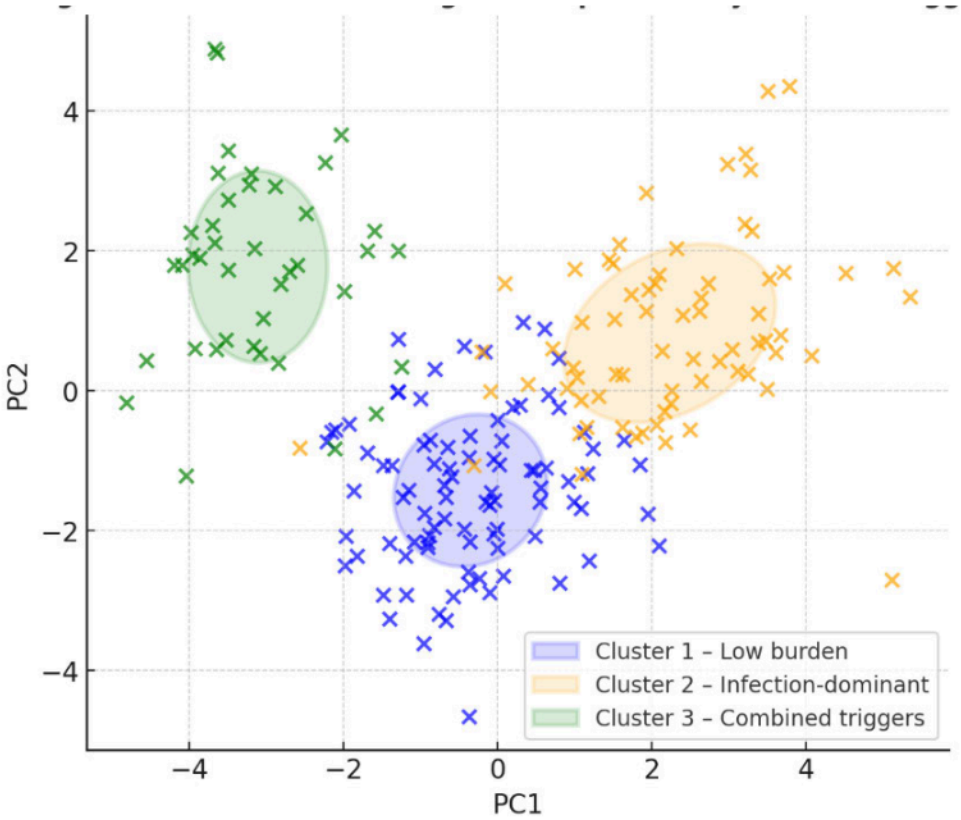

Figure 3. K-means clustering of CKD patients by wartime triggers

Note: Source: authors' analysis; K-means clustering performed in Python (scikit-learn) on wartime trigger indices (monitoring interruption, MIS, PSS, IBI).

The proposed causal framework of *multimorbid-stress-induced CKD dynamics* is summarized in Figure 4, illustrating the pathway from wartime triggers through mediators (eGFR decline, complications) to adverse outcomes (progression and mortality).

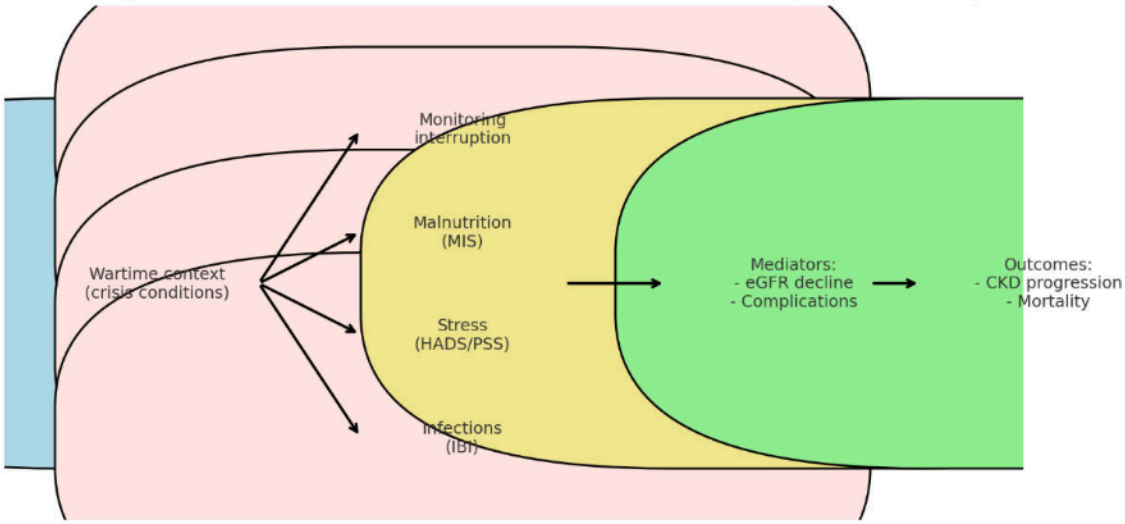

Figure 4. Conceptual framework of multimorbid-stress-induced CKD dynamics during wartime
Note: Source: authors' conceptual diagram; created in Python (matplotlib).

DISCUSSION

Our findings demonstrate that four wartime-specific triggers—monitoring interruption, malnutrition, stress, and infection burden—were independently associated with CKD progression, complications, and mortality. These results align with growing evidence that both traditional and non-traditional risk factors drive adverse outcomes in CKD. For instance, deterioration of the estimated glomerular filtration rate (eGFR) slope remains a powerful predictor of mortality in chronic disease populations [6], while malnutrition and sarcopenia are recognized as independent drivers of CKD progression and death [5, 10]. Likewise, stress-induced metabolic dysregulation, including hyperglycemia, has been shown to worsen prognosis in nephropathy [12, 17].

Among all triggers, infection burden emerged as the most potent predictor of mortality. This finding corresponds to prior reports that recurrent infections and heightened inflammatory status markedly elevate mortality risk in CKD [3, 4, 7]. The impact of cardiovascular comorbidity also remained substantial, consistent with long-term trends in CKD-related cardiovascular mortality [2, 14–16, 18, 19].

Cluster analysis revealed three distinct patient profiles under wartime conditions: (i) low trigger burden with favorable prognosis, (ii) infection-dominant with the highest mortality, and (iii) combined triggers with moderate-to-high risk. This heterogeneity reflects the interplay of multimorbidity, lifestyle, and social determinants of health described in large population studies [13, 16].

The observed 14% all-cause mortality during the six-month observation period is striking and falls within the range reported in international cohorts, though likely aggravated by crisis conditions [2, 4, 18]. Cardiovascular and infectious deaths predominated, echoing global data on the dual burden of atherosclerosis and infections in CKD populations [3, 15, 19].

Overall, these results support a conceptual model of multimorbid-stress-induced CKD dynamics, underscoring that war and humanitarian crises act as amplifiers of existing risk factors. This paradigm resonates with recent analyses highlighting multimorbidity, oxidative stress, malnutrition, and social vulnerability as major determinants of CKD outcomes [8–10, 13].

Limitations include the retrospective, single-center design, which constrains generalizability. Nonetheless, the integration of survival analysis, regression modeling, and cluster techniques enhances the internal validity of the findings.

Future perspectives. Building on recent advances in CKD prognostication [7, 11, 20], we propose the development of a CKD-WarRisk Score to facilitate early identification of high-risk patients in resource-limited and wartime settings. Such a score could integrate clinical, biochemical, and psychosocial dimensions to enable personalized risk prediction, guide targeted interventions, and optimize allocation of scarce healthcare resources. Beyond its immediate application, this tool may serve as a prototype for crisis-adapted risk models in other chronic diseases.

CONCLUSIONS

1. In patients with CKD stages 1–3 observed during wartime (2022–2025), four independent triggers of adverse outcomes were identified: monitoring interruption, malnutrition, psychosocial stress, and infection burden.
2. Each of these triggers demonstrated a significant association with CKD progression, complications, or mortality. In particular, infection burden was the strongest predictor of all-cause mortality (HR 2.5; 95% CI 1.6–3.8).
3. Cluster analysis revealed three phenotypic patterns of CKD trajectories under wartime conditions: (1) low trigger burden with favorable prognosis, (2) infection-dominant with highest mortality, and (3) combined triggers with moderate-to-high risk of complications.
4. Overall mortality during the observation period reached 14%, with cardiovascular and infectious causes predominating.
5. The proposed conceptual model of multimorbid-stress-induced CKD dynamics provides a novel framework for risk stratification in crisis settings and may serve as the basis for developing a CKD-WarRisk Score to guide patient management under limited healthcare resources.

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Мультиморбідні та стрес-індуковані тригери прогресування та смертності при хронічній хворобі нирок у воєнний час: ретроспективне когортне дослідження

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***Анотація. Вступ. З**бройний конфлікт створює унікальні стресори для ведення хронічних захворювань. Пацієнти з хронічною хворобою нирок (ХХН) є особливо вразливими до переривання лікування, мальнутриції, психоемоційного стресу та інфекційного навантаження. **Мета дослідження**. Визначити воєнні тригери прогресування ХХН, провести кластеризацію пацієнтів за домінуючими факторами ризику та оцінити їхній вплив на клінічні наслідки. **Матеріали та методи**. Проведено ретроспективний аналіз 303 пацієнтів із ХХН 1–3 стадії, яких спостерігали протягом 6 місяців (лютий 2022 р. – вересень 2025 р.). Збирали клінічні дані, індекс коморбідності Чарлсона (CCI), нутритивно-запальний індекс (MIS), шкалу сприйняття стресу (PSS) та індекс інфекційного навантаження (IBI). Пацієнтів було стратифіковано у чотири кластери залежно від домінуючих воєнних тригерів. Для оцінки асоціацій застосовували регресійний аналіз Кокса та ROC-аналіз. **Результати**. Виділено чотири кластери: Кластер A (n=84): часті переривання лікування, високе інфекційне навантаження; Кластер B (n=76): мальнутриція та анемія; Кластер C (n=95): високі показники стресу; Кластер D (n=48): поєднані тригери. Під час спостереження загальна смертність сягнула 14%, переважно у кластерах A та B. Інфекційне навантаження виявилося найсильнішим предиктором смертності (HR = 2,5; 95% ДІ 1,8–3,4). Мальнутриція незалежно асоціювалася з ризиком госпіталізації (HR = 1,9; 95% ДІ 1,3–2,7). Кластер стресу характеризувався підвищеною частотою серцево-судинних подій (HR = 1,6; 95% ДІ 1,1–2,3). **Висновки.** Пацієнти з ХХН, які зазнали воєнних порушень у медичному супроводі, мають чіткі, тригер-специфічні ризики. Запропонована кластерна модель підкреслює необхідність пріоритизації інфекційного контролю, нутритивної підтримки та менеджменту стресу як невідкладних напрямів нефропротекції у гуманітарних кризах.*

***Ключові слова:** хронічна хвороба нирок; прогресування ХХН; смертність; воєнні тригери; мультиморбідність; стрес; мальнутриція; інфекції; ретроспективна когорта.*

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Author contributions

Lidiia D. Denova – data collection and analysis, manuscript writing, literature review on the topic, preparation of the manuscript for publication;

Stella V. Kushnirenko – conceptualization, methodology.

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